

R4DS 02 – Data Visualization

Introduction

"The simple graph has brought more information to the data analyst's mind than any other device." — John Tukey

- **ggplot2** implements a *grammar of graphics*
- Learn one system, apply it everywhere
- Core idea: **map** variables in data to **aesthetic** properties (position, color, size, shape)
- Build plots **layer by layer**

Prerequisites

```
library(tidyverse)
```

```
library(palmerpenguins)  
library(ggthemes)
```

- `palmerpenguins`: the `penguins` dataset
- `ggthemes`: colorblind-safe palettes

First Steps

The penguins Data

penguins

```
#> # A tibble: 344 x 8
#>   species island  bill_length_mm bill_depth_mm
#>   <fct>    <fct>          <dbl>          <dbl>
#> 1 Adelie  Torgersen         39.1           18.7
#> 2 Adelie  Torgersen         39.5           17.4
#> 3 Adelie  Torgersen         40.3            18
#> 4 Adelie  Torgersen          NA            NA
#> 5 Adelie  Torgersen         36.7           19.3
#> ... [output truncated]
```

- 344 observations, 8 variables
- Species: Adelie, Chinstrap, Gentoo
- Key variables: `flipper_length_mm`, `body_mass_g`, `bill_length_mm`, `bill_depth_mm`

- **Variable:** a quantity or property you can measure
- **Value:** the state of a variable when you measure it
- **Observation** (data point): a set of measurements on the same object
- **Tabular data** is *tidy* if each value has its own cell, each variable its own column, each observation its own row

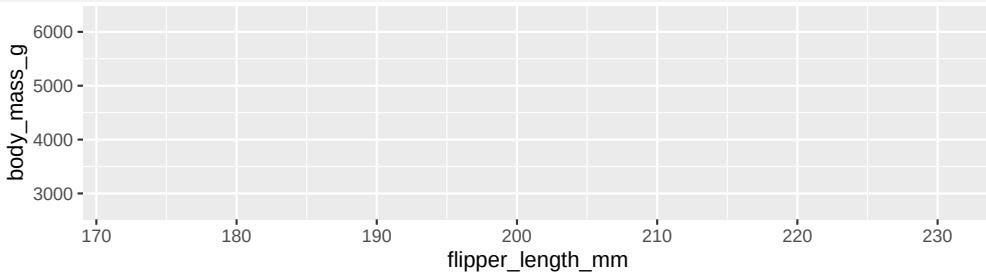
Building a ggplot — Step 1: Data

```
ggplot(data = penguins)
```

`ggplot(data = ...)` creates an empty canvas primed with a dataset.

Building a ggplot — Step 2: Aesthetics

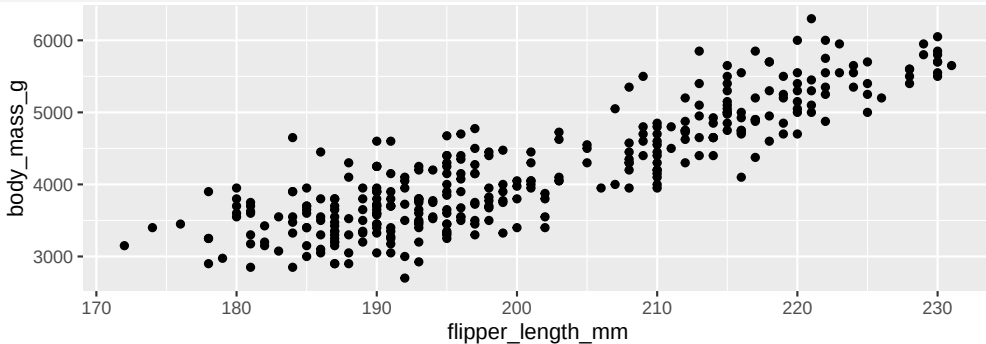
```
ggplot(  
  data = penguins,  
  mapping = aes(x = flipper_length_mm, y = body_mass_g)  
)
```



`mapping = aes(x = ..., y = ...)` maps variables to axes.

Building a ggplot — Step 3: Geom

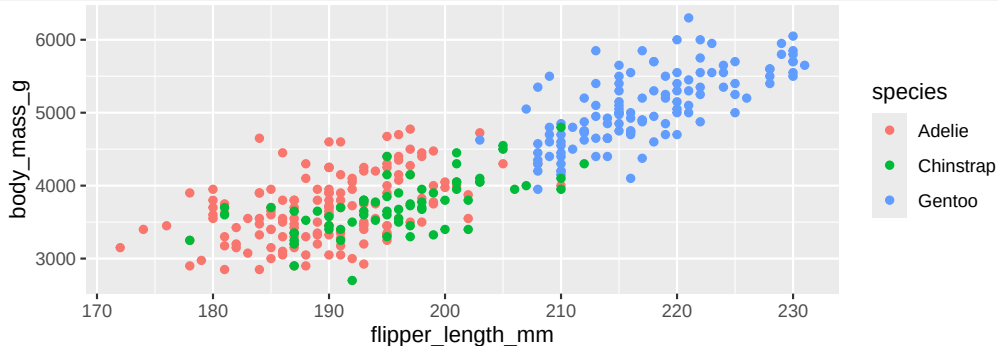
```
ggplot(  
  data = penguins,  
  mapping = aes(x = flipper_length_mm, y = body_mass_g)  
) +  
  geom_point()
```



`geom_point()` adds a **layer** of points → scatterplot.

Adding Color by Species

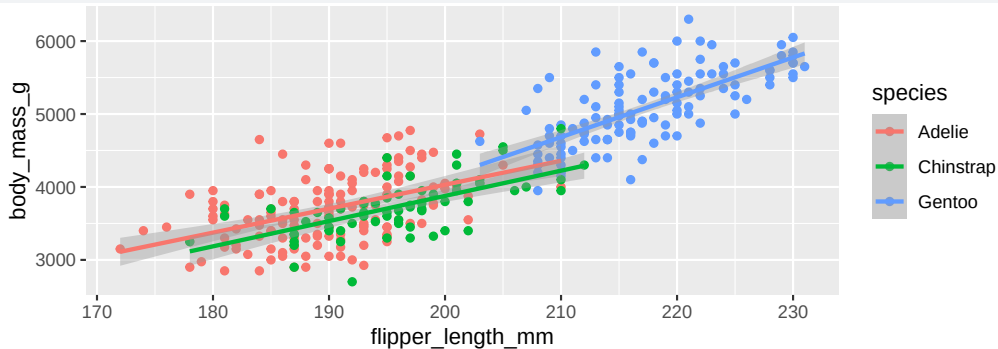
```
ggplot(  
  data = penguins,  
  mapping = aes(x = flipper_length_mm, y = body_mass_g,  
                 color = species)  
) +  
  geom_point()
```



Map a categorical variable to `color` → automatic legend.

Adding a Smooth Line

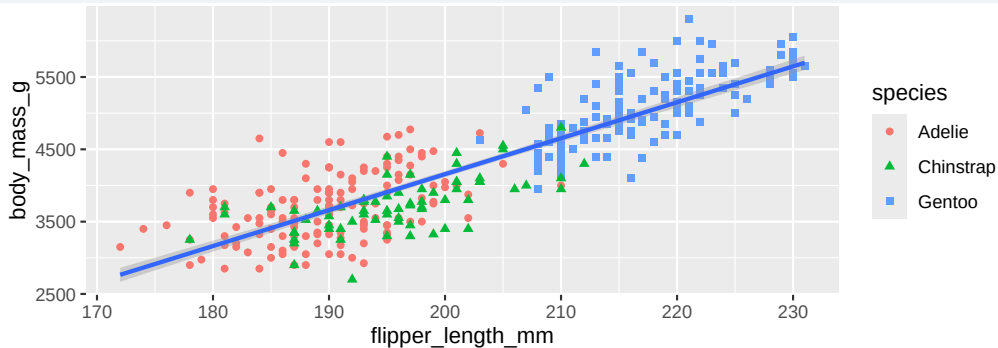
```
ggplot(penguins,  
  aes(x = flipper_length_mm, y = body_mass_g,  
    color = species)) +  
  geom_point() +  
  geom_smooth(method = "lm")
```



`color = species` in `ggplot()` → **both** geoms inherit it → 3 separate lines.

Global vs. Local Aesthetics

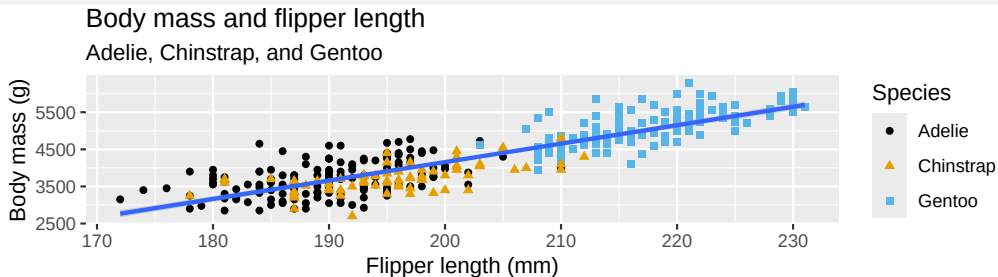
```
ggplot(penguins,  
  aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(aes(color = species, shape = species)) +  
  geom_smooth(method = "lm")
```



Move `color` and `shape` into `geom_point()` → **one** overall line.

The Polished Plot

```
ggplot(penguins,  
  aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(aes(color = species, shape = species)) +  
  geom_smooth(method = "lm") +  
  labs(title = "Body mass and flipper length",  
    subtitle = "Adelie, Chinstrap, and Gentoo",  
    x = "Flipper length (mm)", y = "Body mass (g)",  
    color = "Species", shape = "Species") +  
  scale_color_colorblind()
```



Exercises 2.2.1 (1–5)

- 1 How many rows are in `penguins`? How many columns?
- 2 What does `bill_depth_mm` describe? (Hint: `?penguins`)
- 3 Scatterplot of `bill_depth_mm` vs. `bill_length_mm`. Describe the relationship.
- 4 Scatterplot of `species` vs. `bill_depth_mm`. What might be a better geom?
- 5 Why does this give an error?

```
ggplot(data = penguins) +  
  geom_point()
```

Exercises 2.2.1 (6–10)

- 6 What does `na.rm` do in `geom_point()`? Default value? Use it set to `TRUE`.
- 7 Add caption: “Data come from the palmerpenguins package.” (Hint: `labs()`)
- 8 Recreate: scatterplot with `bill_depth_mm` mapped to color, smooth curve overlaid. Global or local?
- 9 Predict the output, then run:

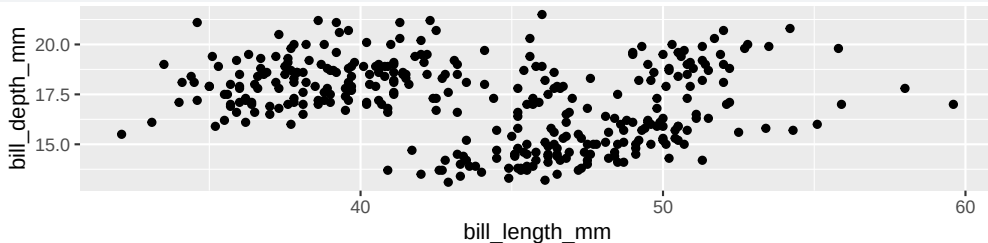
```
ggplot(penguins, aes(x = flipper_length_mm,  
  y = body_mass_g, color = island)) +  
  geom_point() + geom_smooth(se = FALSE)
```

- 10 Will these two plots look different? Why?

Solutions 2.2.1 (1–5)

- 1 344 rows, 8 columns.
- 2 Bill depth in mm — vertical distance from top to bottom of bill.
- 3

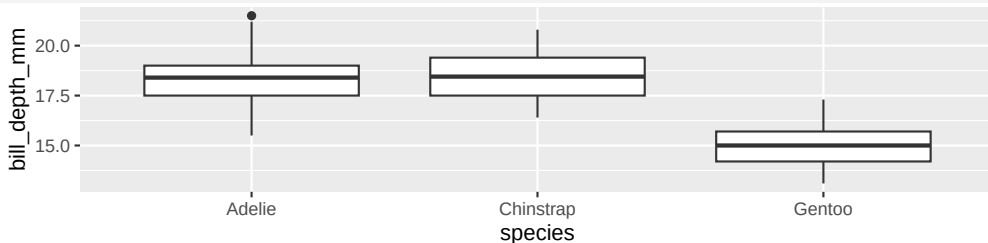
```
ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +  
  geom_point()
```



Negative overall; Simpson's paradox — positive within each species.

4

```
ggplot(penguins, aes(x = species, y = bill_depth_mm)) +  
  geom_boxplot()
```



Categorical $\times$ numerical $\rightarrow$ boxplot is better than scatterplot.

5 `geom_point()` needs `x` and `y` aesthetics. No `mapping` was provided.

Solutions 2.2.1 (6–8)

- ⑥ `na.rm = TRUE` silently removes NAs (default: `FALSE`, which warns).

```
ggplot(penguins, aes(x = flipper_length_mm,  
  y = body_mass_g)) +  
  geom_point(na.rm = TRUE)
```

- ⑦ Add `labs(caption = "Data come from the palmerpenguins package.")`
- ⑧ `bill_depth_mm` mapped to `color` at the **geom level** (so the smooth curve is uncolored):

```
ggplot(penguins, aes(x = flipper_length_mm,  
  y = body_mass_g)) +  
  geom_point(aes(color = bill_depth_mm)) +  
  geom_smooth()
```

- 9 Three smooth curves (one per island, no confidence bands) with colored points.
- 10 **No** — both plots look the same. The first specifies `data` and `mapping` globally; the second repeats them in each geom. Same result either way.

Concise ggplot2 Calls

Dropping Argument Names

Verbose:

```
ggplot(  
  data = penguins,  
  mapping = aes(x = flipper_length_mm, y = body_mass_g)  
) +  
  geom_point()
```

Concise (first two arguments by position):

```
ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point()
```

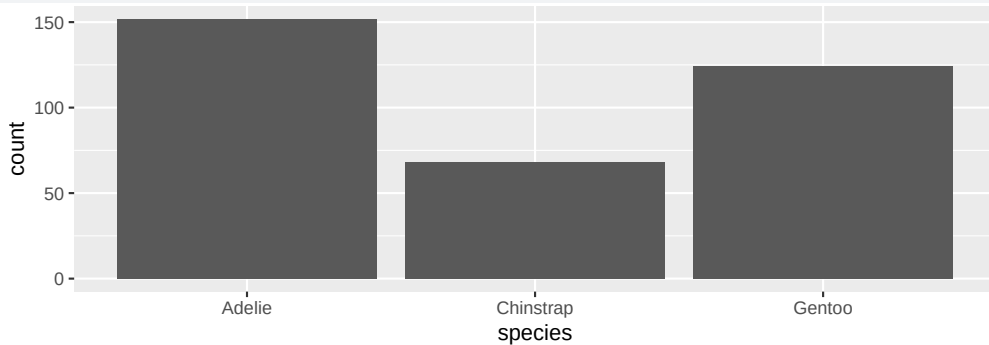
With the pipe:

```
penguins |>  
  ggplot(aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point()
```

Visualizing Distributions

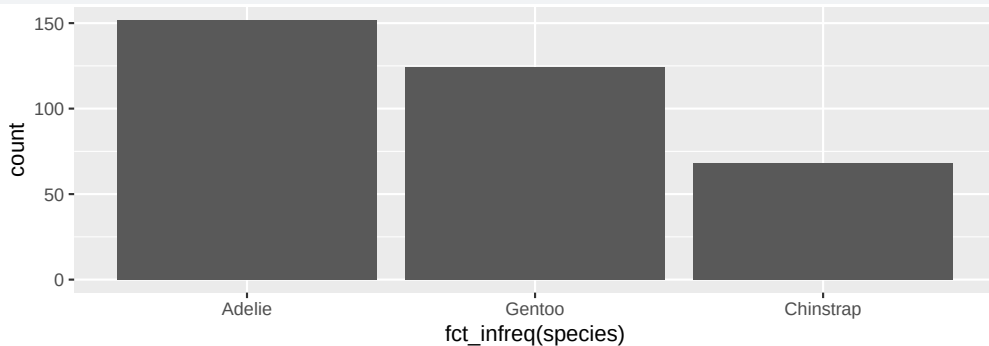
Categorical Variable — Bar Chart

```
ggplot(penguins, aes(x = species)) +  
  geom_bar()
```



Reorder Bars by Frequency

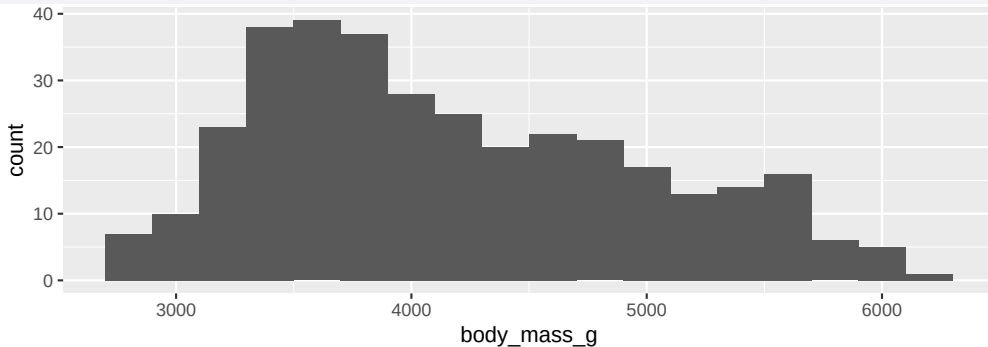
```
ggplot(penguins, aes(x = fct_infreq(species))) +  
  geom_bar()
```



`fct_infreq()` reorders factor levels by decreasing frequency.

Numerical Variable — Histogram

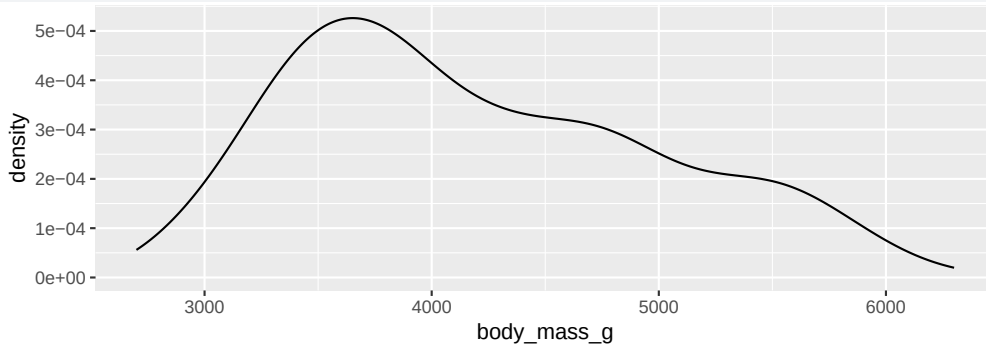
```
ggplot(penguins, aes(x = body_mass_g)) +  
  geom_histogram(binwidth = 200)
```



`binwidth` controls bin width (in units of `x`). Always try several values.

Numerical Variable — Density Plot

```
ggplot(penguins, aes(x = body_mass_g)) +  
  geom_density()
```



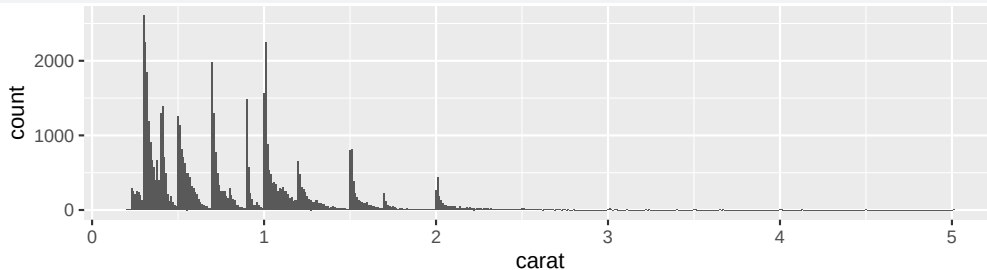
Smoothed version of a histogram — good for seeing overall shape.

- 1 Bar plot of `species` with `species` on the *y*-axis. How is it different?
- 2 `color = "red"` vs. `fill = "red"` in `geom_bar()`. Which is more useful?
- 3 What does the `bins` argument in `geom_histogram()` do?
- 4 Histogram of `carat` in `diamonds`. Which binwidth reveals the most interesting patterns?

Solutions 2.4.3

- ① Horizontal bars instead of vertical.
- ② `fill` changes the interior color; `color` only the outline. `fill` is more useful.
- ③ `bins` sets the number of bins (default 30). Alternative to `binwidth`.
- ④

```
ggplot(diamonds, aes(x = carat)) +  
  geom_histogram(binwidth = 0.01)
```

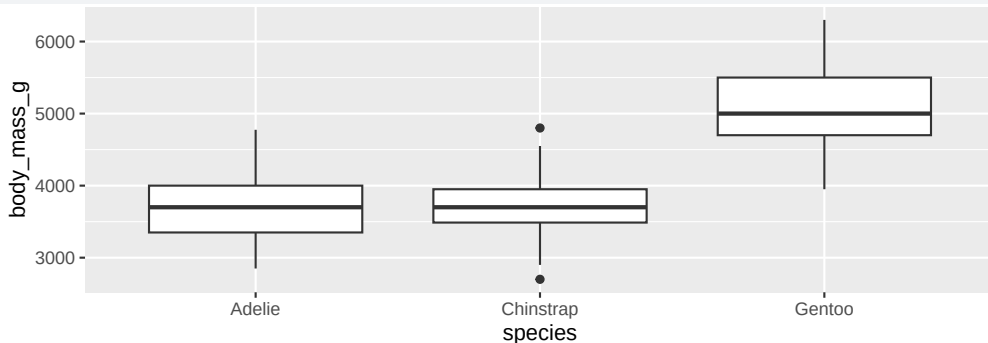


Spikes at “round” values (0.5, 1.0, 1.5, 2.0).

Visualizing Relationships

Numerical + Categorical — Boxplot

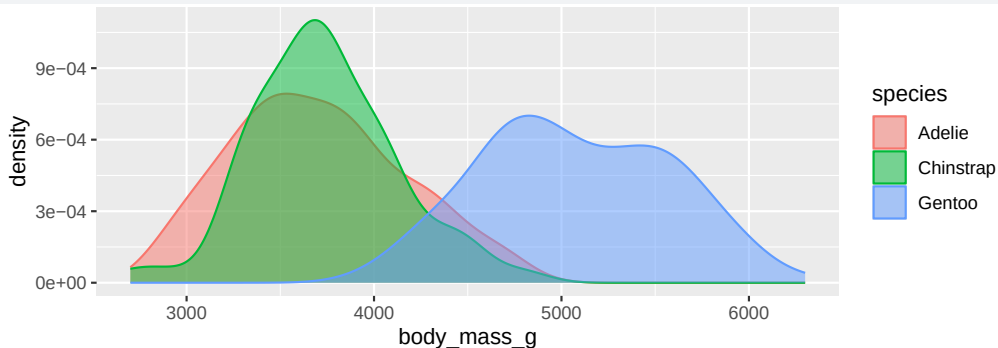
```
ggplot(penguins, aes(x = species, y = body_mass_g)) +  
  geom_boxplot()
```



Box = IQR (25th–75th percentile); line = median; whiskers = $1.5 \times \text{IQR}$; dots = outliers.

Numerical + Categorical — Density

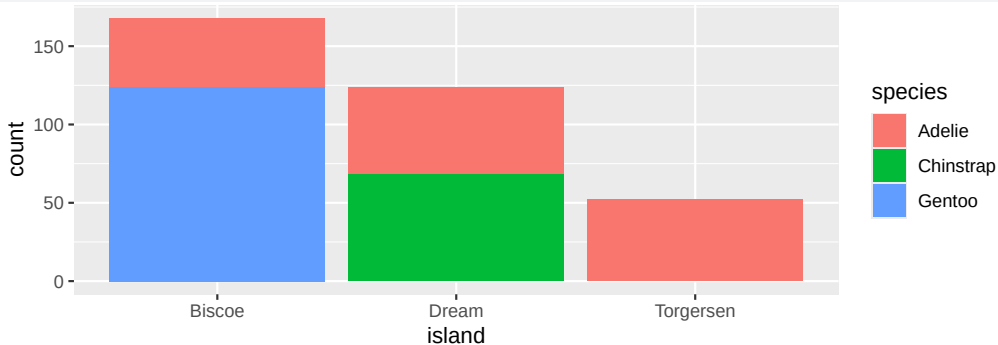
```
ggplot(penguins, aes(x = body_mass_g,  
                    color = species, fill = species)) +  
  geom_density(alpha = 0.5)
```



`alpha = 0.5` adds transparency. We **map** variables to aesthetics; we **set** fixed values.

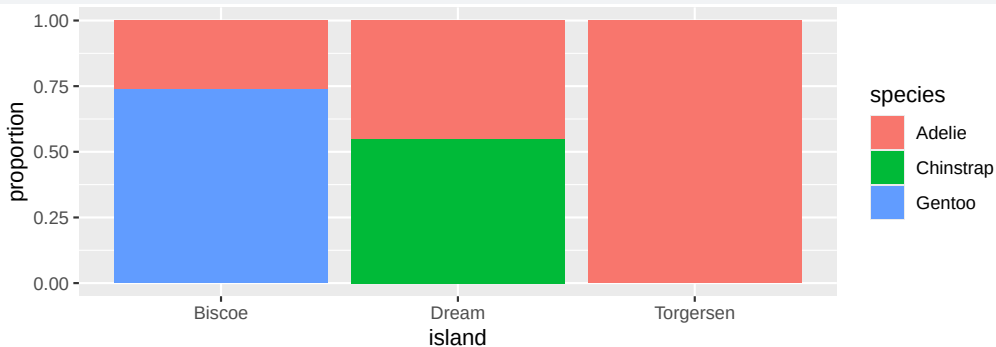
Two Categorical Variables — Stacked Bar

```
ggplot(penguins, aes(x = island, fill = species)) +  
  geom_bar()
```



Relative Frequency Bar

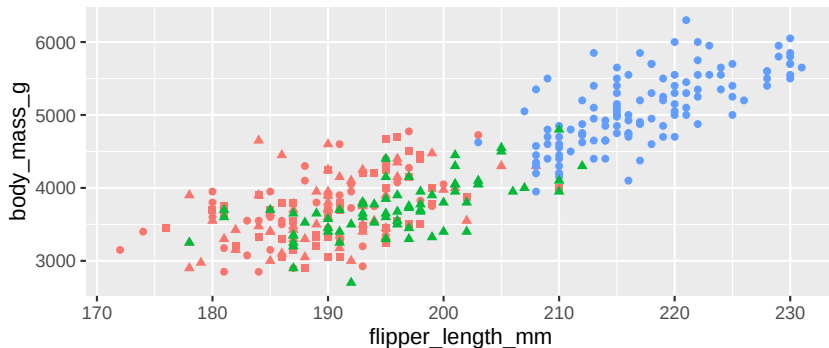
```
ggplot(penguins, aes(x = island, fill = species)) +  
  geom_bar(position = "fill") +  
  labs(y = "proportion")
```



`position = "fill"` scales each bar to 100% for comparing proportions.

Three or More Variables — Extra Aesthetics

```
ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(aes(color = species, shape = island))
```



species

- Adelie
- Chinstrap
- Gentoo

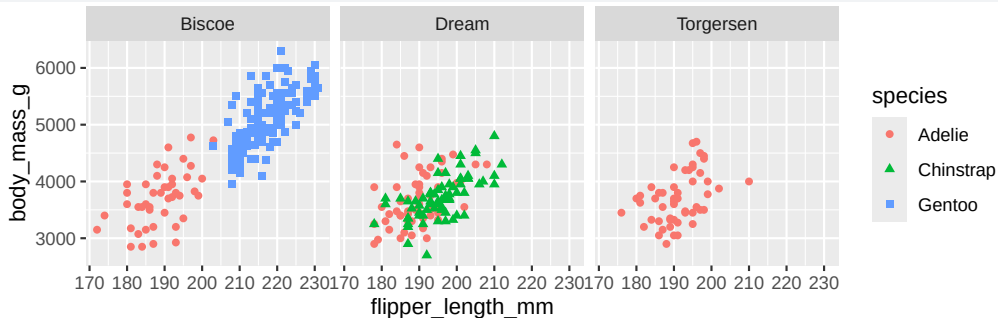
island

- Biscoe
- Dream
- Torgersen

Gets cluttered fast — consider **facets** instead.

Faceting

```
ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point(aes(color = species, shape = species)) +  
  facet_wrap(~island)
```



`facet_wrap(̃var)` splits the plot into subplots by a categorical variable.

Exercises 2.5.5 (1–4)

- 1 Which variables in `mpg` are categorical? Numerical? How can you tell from `mpg`?
- 2 Scatterplot of `hwy` vs. `displ`. Map a third variable to `color`, then `size`, then both, then `shape`. Categorical vs. numerical differences?
- 3 What happens if you map a third variable to `linewidth`?
- 4 What happens if you map the same variable to multiple aesthetics?

Exercises 2.5.5 (5–7)

- 5 Scatterplot of `bill_depth_mm` vs. `bill_length_mm`, colored by `species`. What does it reveal? What about faceting?
- 6 Why does this yield two separate legends? Fix it.

```
ggplot(penguins, aes(x = bill_length_mm,  
  y = bill_depth_mm, color = species,  
  shape = species)) +  
  geom_point() + labs(color = "Species")
```

- 7 Create two stacked bar plots (island by species, species by island). Which question does each answer?

- ① Categorical: `manufacturer`, `model`, `trans`, `drv`, `fl`, `class`. Numerical: `displ`, `year`, `cyl`, `cty`, `hwy`. The tibble shows `<chr>` vs. `<dbl>/<int>`.
- ② Numerical variables produce a continuous color gradient or continuous size; categorical produce discrete colors/shapes. `shape` cannot be used with numerical variables.
- ③ `linewidth` has no effect on `geom_point()` — it's for lines.
- ④ Redundant encoding; ggplot2 may combine legends.

- 5 Without color: negative trend (Simpson's paradox). With color: **positive** within each species. Faceting shows the same.
- 6 `labs(color = "Species")` sets only the color legend label. Fix: also set `labs(shape = "Species")` or use matching labels.
- 7 First (`x = island, fill = species`): "What is the species composition of each island?" Second (`x = species, fill = island`): "Which islands does each species inhabit?"

Saving Plots

```
ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g)) +  
  geom_point()  
ggsave(filename = "penguin-plot.png")
```

- Saves the **most recently displayed** plot by default
- Specify **width** and **height** for reproducibility
- File extension determines format: **.png**, **.pdf**, **.svg**, etc.

- ❶ Which plot is saved as `mpg-plot.png`? Why?

```
ggplot(mpg, aes(x = class)) + geom_bar()  
ggplot(mpg, aes(x = cty, y = hwy)) + geom_point()  
ggsave("mpg-plot.png")
```

- ❷ Change the code to save as PDF instead. How can you find supported formats?

- 1 The **scatterplot** (`cty` vs. `hwy`) is saved, because `ggsave()` defaults to the **last displayed plot**.
- 2 Change `"mpg-plot.png"` to `"mpg-plot.pdf"`. Run `?ggsave` to see supported formats (or check the file extension).

Common Problems

Common mistakes:

- **Misplaced +**: must be at the **end** of the line, not the start

WRONG:

```
ggplot(data = mpg)  
+ geom_point(aes(x = displ, y = hwy))
```

RIGHT:

```
ggplot(data = mpg) +  
  geom_point(aes(x = displ, y = hwy))
```

- **Missing aesthetic**: error says “stat requires ... aesthetic”
- **Unmatched quotes/parens**: console shows + prompt — press **Escape**
- **Google the error message**: someone else has had the same problem